

VPC

1. What is Amazon Virtual Private Cloud (Amazon VPC), and why is it important in AWS networking?
2. What is the primary difference between a public subnet and a private subnet in a VPC?
3. How do you connect a VPC to an on-premises data center, and what are the options available for this connection?
4. Explain the purpose of Amazon VPC peering and its use cases.
5. What is the significance of route tables in a VPC, and how do you control traffic routing between subnets?
6. What are VPC Endpoints, and how do they enhance security and reduce data transfer costs for certain AWS services?
7. Explain the use of a Bastion Host (Jump Host) in a VPC for secure remote access to instances.
8. What is Direct Connect, and how does it provide dedicated network connectivity between an on-premises data center and an AWS VPC?
9. Describe the concept of VPC Flow Logs and their benefits for network monitoring and troubleshooting.
10. What is AWS Transit Gateway, and how does it simplify network connectivity and management in complex VPC architectures?

11. Explain the use of AWS PrivateLink for securely accessing AWS services over private connections within a VPC.
12. What are some best practices for designing VPC architectures that are highly available, fault-tolerant, and scalable?
13. Give examples of scenarios where you would use VPC peering, VPC endpoints, or Direct Connect to enhance network connectivity.
14. Discuss strategies for managing and optimizing VPC resources, including IP address allocation, subnet sizing, and route table design.
15. What are the considerations when setting up VPCs in a multi-region or global configuration for disaster recovery or load balancing?